CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of Bachelor of Technology (B.Tech.)
University Examination May2011

EE 206 Digital Electronics and Microprocessors

Date: 13/05/2011, Friday **Time: 10:00 a.m. to 1:00 p.m.** **Maximum Marks: 70**

Instructions:
1. Write both the sections in separate answer sheets.
2. Clearly mention the assumptions if any.

SECTION-I

			Marks
Q.1	(a)	Write the Boolean algebraic laws.	2
	(b)	Do as directed. I) Multiply 1101_2 by 110_2 II) Convert 4057.06_8 to decimal. III) Convert 1011011011_2 to hexadecimal.	6
	(c)	I) Each term in the standard SOP form is called a ______. (a) minterm (b)maxterm (c) Don't care (d)literal II) The code used for labeling the cells of a k-map is ______. (a)8-4-2-1 binary (b)hexadecimal (c)gray (d) octal	2
Q.2	(a)	What are the merits and demerits of C-MOS?	4
	(b)	Reduce using mapping the expression f=Σm(0, 1, 3, 4, 5, 6, 7, 13, 15) and implement the real minimal expression in universal logic.	6
		OR	
Q.2	(a)	With the help of a neat diagram explain the working of a 2-input TTL NAND Gate.	4
	(b)	Simplify the following expression once by considering the don't care condition and once by ignoring the don't care condition. Y=Σm(1,4,8,12,13,15) + d(3,14).	6
Q.3	(a)	Explain a BCD to 7-segment decoder using 4-line to 16-line decoder with diagram and truth table.	4
	(b)	Build a full adder from half adder circuit and also draw its truth table.	4
	(c)	With a neat diagram explain the operation of a 4-bit SISO register. Draw the timing diagram and give its truth table.	4
	(d)	What is counter? What are its types? What are the applications of a counter?	3
		OR	
Q.3	(a)	Implement the following Boolean expression using 8:1 multiplexer. f(A,B,C,D)= A'B'D'+ABC+B'CD+A'CD	4
	(b)	Explain with diagram the working of D-type flip flop. Give its truth table.	4
	(c)	Explain working of 2-bit asynchronous binary up counter using T-Flip flop with suitable circuit and timing diagram.	4
	(d)	Explain 3 to 8-line decoder.	3

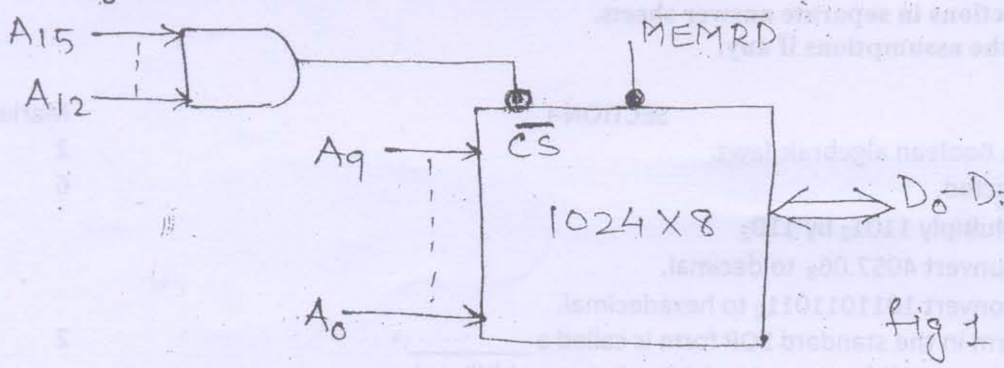
SECTION-II

Q.4 (a) Explain the following terms in brief. 5

1. Microprocessor 2.Compiler
3. Address bus 4.Data bus
5. Flag Register

(b) Explain the execution of CALL instruction in 8085 microprocessor. 6

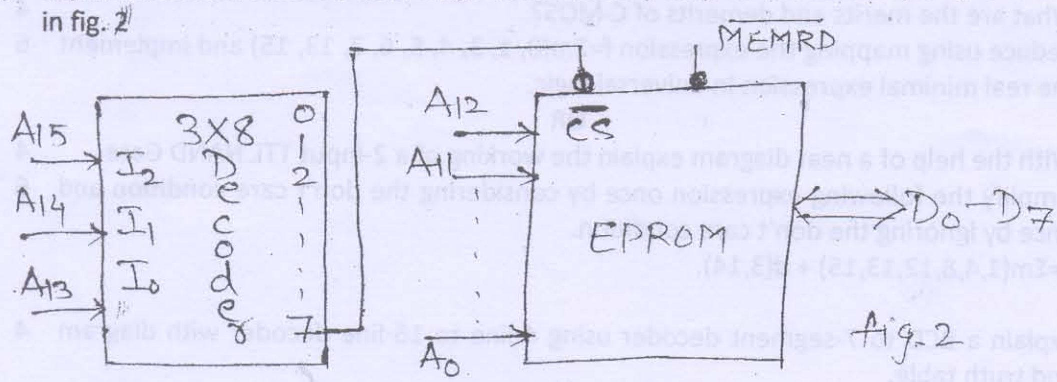
Q.5 (a) Explain partial address decoding. Find memory map of the memory chip shown in fig. 1 6



(b) Draw the timing diagram for LDA 2500h instruction. 6

OR

Q.5 (a) Explain absolute address decoding. Find memory map of the memory chip shown in fig. 2 6



Line 7 is low when $A_{15} \rightarrow A_{13}$ are high

(b) Draw the timing diagram for STA 2400h instruction. 6

Q.6 Write 8085 microprocessor based assembly program for any three of the following. Explain the program logic in brief.. 12

(a) To sort the given array of data in descending order.

(b) Find the total number of bytes having MSB and LSB as 1 from the given array of data 8-bit data.

(c) Delay subroutine to have delay of 10 ms with external clock frequency of 3 MHz.

(d) Discard all the negative numbers and add rest of the data bytes from the given array of 8-bit data. Store the result (sum and carry) in two consecutive memory locations.

(e) Copy the block of data starting from 2300 H to the memory location 2400 H.

(f) Discard all odd numbers from the array of 8-bit data and add rest of the numbers.